

AI Action Plan - Advancing U.S. Leadership in AI

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Submitted by email to ostp-ai-rfi@nitrd.gov

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan

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Introduction

America, as a global leader in AI innovation, has the unique opportunity and responsibility to establish the standards for a secure, ethical, and safe AI future. In support of the OSTP's efforts to define the policy actions needed to sustain and enhance America's AI dominance, I present an action plan.

This AI action plan centers on three pivotal pillars designed to propel America's leadership in AI innovation. It calls for *enhancing AI security* by developing and implementing robust measures to safeguard systems from cyberattacks and malicious use throughout their lifecycle. Additionally, it emphasizes *strengthening AI data privacy* by establishing clear guidelines and regulations that ensure the ethical and responsible use of data, thereby protecting privacy and security. Finally, the plan *promotes AI safety* through the development and deployment of systems that are safe, reliable, and aligned with human values, effectively minimizing potential risks and unintended consequences.

Key Initiatives

To achieve these goals, this plan proposes the following key initiatives, prioritized based on their potential impact and urgency:

AI Security

- Promote research on AI security technologies

Invest in research and development of AI-specific security solutions, such as defenses against data breaches, model poisoning, backdoor attacks and adversarial attacks¹.

- Establish a framework for AI risk management

Develop a framework to assess and manage AI risks, including cybersecurity threats, data privacy breaches, and safety concerns. The framework should provide guidance on risk assessment, mitigation, and best practices for secure AI development.

Data Privacy and Governance

- Establish guidelines for responsible AI

Develop clear guidelines for the ethical and responsible use of AI, ensuring fairness, transparency, and accountability. This includes addressing concerns about unauthorized data use, biometric data concerns, and covert data collection².

Guidelines should ensure that AI systems do not perpetuate or amplify existing societal biases.

- Promote data security best practices

Encourage the adoption of data security best practices, such as encryption, access controls, and data anonymization, to protect sensitive information used in AI systems. This includes conducting regular audits to ensure compliance with privacy regulations.

- AI Legislation Reformation

Analyze the current AI legislation landscape in the US and globally, including examples like the GDPR, EU AI Act, and Algorithmic Accountability Act. Identify areas where existing legislation may need to be updated or strengthened to address the unique challenges of AI, such as algorithmic transparency, data protection, and liability for AI-related harms.

Cybersecurity Diplomacy

- Promote international standards

America can play a critical role in leading international partnerships to develop and

¹ Deep Learning Model Security: Threats and Defenses https://ui.adsabs.harvard.edu/link_gateway/2024arXiv241208969W/doi:10.48550/arXiv.2412.08969

² The ethics of facial recognition technologies, surveillance, and accountability in an age of artificial intelligence: a comparative analysis of US, EU, and UK regulatory frameworks. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8320316/>

promote clear standards and guidelines for AI security and privacy. This includes harmonizing regulations across borders to ensure interoperability and prevent regulatory fragmentation.

- Collaborate on threat intelligence and response

Establish mechanisms for sharing AI threat intelligence and collaborating on incident response with international partners to effectively address cross-border AI security threats. This includes developing secure communication channels and protocols for sharing information about AI-related vulnerabilities and attacks³.

Public Literacy

- Public awareness and education

Conduct public awareness campaigns and engage in open dialogues to educate the public about AI security and privacy risks and foster informed discussions about responsible AI development. This includes creating educational resources, hosting public forums, and engaging with civil society organizations to promote public understanding of AI.

Timeline

Under the forward-thinking leadership of President Trump and this Administration, this four-year roadmap outlines how to strengthen AI security, data privacy, global cybersecurity diplomacy, and public literacy. Each year focuses on key milestones that build on prior progress,

Year 1 (2025–2026)

- AI Security
 - Launch research projects on potential security threats and defense mechanisms in AI.
 - Start developing prototypes for AI-driven cybersecurity tools.
- Data Privacy and Governance
 - Form a specialized task force dedicated to AI data privacy.
 - Draft guidelines for responsible AI data usage, emphasizing transparency, fairness and accountability.
- Cybersecurity Diplomacy

³ Cyber diplomacy: defining the opportunities for cybersecurity and risks from Artificial Intelligence, IoT, Blockchains, and Quantum Computing

<https://www.tandfonline.com/doi/full/10.1080/23742917.2024.2312671>

- Begin discussions with international partners on AI standards and security goals.
- Lay the groundwork for collaborative threat intelligence sharing.
- Public Literacy
 - Identify key audiences and conduct surveys to determine their most pressing concerns about AI.
 - Develop introductory, easy-to-understand educational materials.
 - Partner with civil society, schools, libraries, and local governments to create a network for disseminating these resources.

Year 2 (2026–2027)

- AI Security
 - Pilot AI-powered cybersecurity tools in real-world settings to measure effectiveness.
- Data Privacy and Governance
 - Test data privacy frameworks within a closed group of hand-picked organizations.
 - Use pilot results to refine and prepare broader privacy regulations.
- Cybersecurity Diplomacy
 - Actively participate in international forums to shape AI standards.
 - Share early successes and lessons learned to encourage global consensus.
- Public Literacy
 - Integrate AI security and privacy lessons into school and university curricula, spanning both STEM and social sciences.
 - Work with libraries and community centers to host educational sessions for diverse audiences.

Year 3 (2027–2028)

- AI Security
 - Roll out AI security solutions on a larger scale, based on successful Year 2 pilots.
 - Continue enhancing defenses against emerging threats.
- Data Privacy and Governance
 - Finalize and adopt data privacy regulations industry-wide.
 - Conduct regular audits to ensure compliance and address new challenges.
- Cybersecurity Diplomacy

- Establish joint research programs on AI security with international partners.
- Develop channels for ongoing intelligence-sharing to address cross-border threats.
- Public Literacy
 - Collaborate with experts and community leaders to produce culturally and linguistically relevant materials.
 - Introduce clear credentialing that rewards individuals and organizations for completing AI literacy modules, boosting motivation for continued learning.

Year 4 (2028 and Beyond)

- AI Security
 - Continuously evaluate and refine AI security measures to stay ahead of evolving threats.
 - Maintain a cycle of updates and testing to ensure resilience.
- Data Privacy and Governance
 - Monitor and enforce data privacy regulations across all sectors.
 - Adapt guidelines as AI applications expand or transform.
- Cybersecurity Diplomacy
 - Formalize frameworks for sharing real-time threat intelligence globally.
 - Strengthen ties with international partners to develop unified standards.
- Public Literacy
 - Involve newly trained community members and leaders in policy discussions at both local and national levels.
 - Encourage public insight and feedback to continually improve AI governance.

Metrics

Measuring success is crucial for continuous improvement. Below are key performance indicators (KPIs) aligned with each initiative:

- Adoption of Secure AI Development Practices
 - Track and audit robust security protocols integrated into AI project lifecycles.
 - Measure risk mitigation effectiveness before public release.
- Reduction in AI-Related Cyber Incidents
 - Monitor the frequency and severity of AI-based attacks.
 - Evaluate response times and overall impact on national security and

consumer trust.

- Increased International Collaboration
 - Count the number of cross-border research initiatives, forums, and threat intelligence exchanges.
 - Track progress toward harmonized global standards for AI security.
- Improved Public Awareness
 - Assess public understanding of AI security and privacy through surveys, feedback, and participation in training programs.
 - Gauge adoption of safe AI practices by individuals, schools, and organizations.

Resourcing the AI Action Plan

Allocating resources to the following areas is essential for the success of this multi-year plan.

- **Funding:** Allocate sufficient funding for
 - Research and Development
 - Workforce Development, and the
 - Establishment of necessary infrastructure.

This will involve securing funding from government agencies, private sector partnerships, and research grants.

- **Public-Private Talent:** Recruit and train skilled personnel in AI security, data privacy, and AI safety to support the implementation of this plan. This includes cybersecurity experts, AI ethicists, data scientists, and legal professionals with expertise in AI. Recruitment efforts should focus on attracting top talent from academia, industry, and government. Create rotational programs between,
 - National labs and tech companies
 - Regulatory agencies and startups
 - Military cyber units⁴ and cloud providers
- **Infrastructure:** Invest in the necessary infrastructure, such as secure data centers, testing facilities, and collaborative platforms, to support secure and privacy-preserving AI development and deployment.

Workforce Development

To effectively implement this AI Action Plan, a skilled workforce capable of addressing AI

⁴ Defense Digital Service <https://www.dds.mil/work>

security and privacy challenges is essential. The following strategies will be employed to develop this workforce:

- **Education and Training:** Invest in educational programs and training initiatives to equip individuals with the necessary skills in AI security, privacy, and ethics. This includes developing specialized curricula for universities and vocational schools, as well as providing online courses and certifications.
- **Upskilling and Reskilling:** Support upskilling and reskilling programs for existing professionals in cybersecurity, data science, and related fields to enhance their knowledge of AI-specific security and privacy challenges.
- **Recruitment and Retention:** Implement strategies to recruit and retain top talent in AI security and privacy, including offering competitive salaries, research opportunities, and professional development programs.
- **Immigration Reforms:** To accelerate AI development in the US, the country should attract and retain global AI talent by simplifying immigration and offering incentives.

Risks and Challenges

Implementing this AI Action Plan may encounter the following risks and challenges:

- **The Evidence Dilemma:** Policymakers face an "evidence dilemma"⁵ in AI policymaking, where AI capabilities are advancing faster than our ability to fully understand their risks and benefits.
- **Lack of awareness and expertise:** A shortage of skilled personnel and a lack of awareness among developers and policymakers may hinder the effective implementation of this plan.
- **Balancing innovation with security:** Finding the right balance between fostering AI innovation and ensuring security and privacy requires a nuanced approach to AI governance that promotes responsible innovation while safeguarding security and privacy.
- **International cooperation:** Achieving international cooperation and harmonization of AI security standards may be challenging due to diverse regulatory approaches and existing geopolitical factors.
- **Environmental Impact of AI:** The increasing adoption of AI technologies requires enormous amounts of computing resources. This has significant implications for energy consumption. Addressing this challenge requires research and development of more energy-efficient AI systems and a focus on sustainable practices in AI

⁵ Global Insights on AI Safety: A Citizen's Perspective on the Landmark International Report

<https://moderndiplomacy.eu/2025/02/06/global-insights-on-ai-safety-a-citizens-perspective-on-the-landmark-international-report/>

development and deployment.

Benefits of Robust AI Security and Privacy

A robust AI security and privacy will bring significant benefits and opportunities:

- **Increased trust and confidence in AI:** Enhanced security and privacy measures will foster greater trust and confidence in AI technologies, encouraging wider adoption and responsible innovation. This will create a more favorable environment for AI development and deployment, leading to economic growth and societal benefits.
- **Economic growth and competitiveness:** A secure and trustworthy AI ecosystem will promote economic growth and competitiveness by enabling the development and deployment of innovative AI solutions across various sectors. This includes applications in healthcare, finance, manufacturing, and transportation, among others.
- **Improved national security:** Robust AI security will strengthen national security by protecting critical infrastructure and mitigating the risks of malicious use of AI. This includes protecting against AI-powered cyberattacks and ensuring the responsible use of AI in defense and intelligence applications.
- **Enhanced societal well-being:** Responsible AI will contribute to societal well-being by ensuring fairness, transparency, and accountability in AI systems. This includes mitigating the risks of bias and discrimination, promoting ethical AI development, and ensuring that AI is used to make humans more humane.

Conclusion

Under President Trump's bold and visionary administration, America can retain and enhance its AI dominance. The initiatives proposed in this action plan aim to amplify the nation's strengths while mitigating emerging risks.

The path forward requires a thoughtful balance. Although robust security and privacy measures are critical, overregulation could stifle the very innovation that drives AI progress. A measured approach, one that nurtures a dynamic, trustworthy AI ecosystem where security and innovation thrive side by side is essential. International collaboration will also be pivotal, allowing nations to share knowledge, harmonize standards, and collectively tackle the shared challenges of AI security. Ultimately, this plan reinforces America's position as a global AI leader that serves the common good, empowers individuals, and bolsters national security while upholding human values.